

CS 660: Data Mining For Decision Making

Lecture 9 (Week 3)

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Scaling the Heights

Decision Tree Classifier on Missing Value Attributes

NAME of training pattern	Attributes			Class
	HABIT	EATS	FOOTWEAR	
T1	Gabby	Baked	Clogs	Student
T2	Gabby	Roasted	Sandals	Professor
T3	Gabby	Baked	Sandals	Student
T4	Quiet	Fried	Sandals	Professor
T5	Gabby	Fried	—	Student
T6	Quiet	Baked	Sandals	Student
T7	Gabby	Fried	Sandals	Professor
T8	Quiet	Fried	Clogs	Student

Theory for Missing Value Attributes

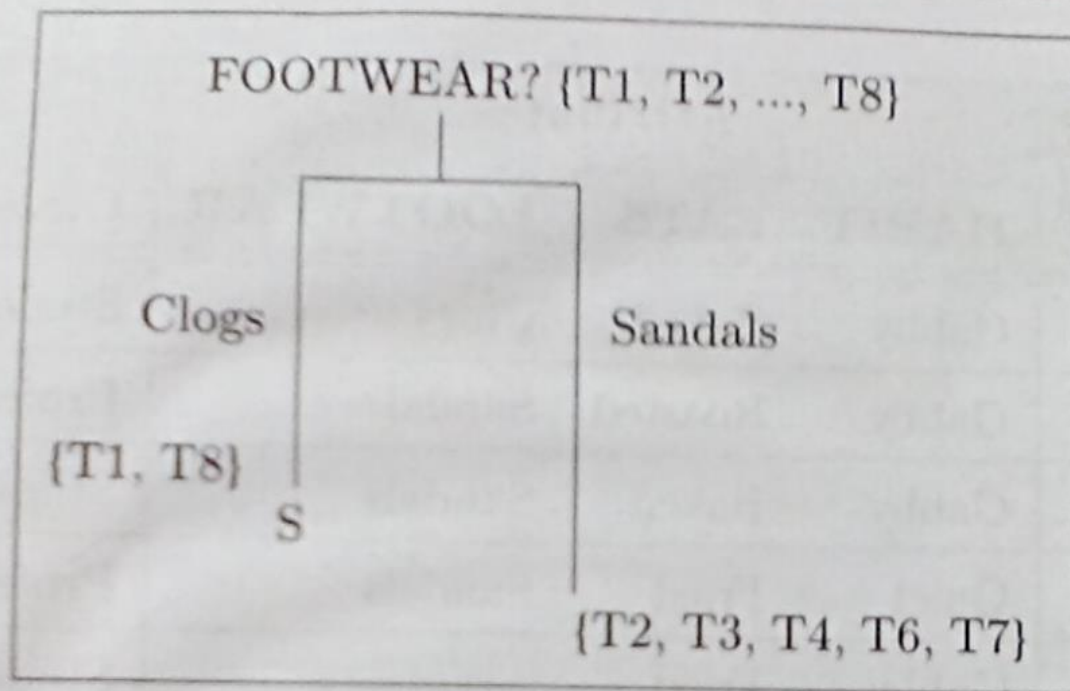
Let $C_1, C_2, \dots, C_m$ be the $m \geq 1$ classes to which the training patterns belong. Suppose V_0 , a subset of the training set V , is associated with a node y_0 of a decision tree we are building. We want to evaluate the ratio of information gain for attribute A at node y_0 , but the value of attribute A is missing for some of the patterns in V_0 .

If attribute A has $n > 1$ possible values, it will split V_0 into sets $V_1, V_2, \dots, V_n$. Those patterns in V_0 whose value of attribute A is missing will not be in any of the sets V_1 to V_n . We will temporarily put such patterns in a set V_{n+1} . For $1 \leq k \leq m$ and $0 \leq i \leq n$, let $Y(k, i)$ be the number of training patterns of class C_k in the set V_i . For $0 \leq i \leq (n + 1)$, let $Z(i)$ be the number of patterns in V_i . Moreover, let

$$Z'(0) = Z(0) - Z(n + 1)$$

Out of $Z(0)$ patterns in set $V(0)$, we thus have $Z'(0)$ patterns for which the value of attribute A is known. Proceeding as we did for Eqn (2.1) in

Not Considering the Missing Value Training Pattern in the Decision Tree



Examining FOOTWEAR at the root of a decision tree to be built for the professor-student training set given in Fig. 3.1, in which the value of FOOTWEAR is missing for pattern T5. Hence, T5 is not associated with either of the children of the root. Both the patterns associated with the left child of the root belong to the class student, and hence the node becomes a leaf node with label S. Of the patterns associated with the right child of the root, patterns T2, T4, and T7 belong to the class professor; and patterns T3 and T6 belong to the class student.

Theory of Missing Value Attributes

$$I(V_0) = - \sum_{k=1}^m \frac{Y(k, 0)}{Z'(0)} \log \frac{Y(k, 0)}{Z'(0)}$$

$$I_A(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z'(0)} \sum_{k=1}^m \frac{Y(k, j)}{Z(j)} \log \frac{Y(k, j)}{Z(j)}$$

$$g_A(V_0) = \frac{Z'(0)}{Z(0)} [I(V_0) - I_A(V_0)]$$

$$S_A(V_0) = - \sum_{j=1}^{n+1} \frac{Z(j)}{Z(0)} \log \frac{Z(j)}{Z(0)}$$

$$G_A(V_0) = \frac{g_A(V_0)}{S_A(V_0)}$$

Calculation of $G_{\text{footwear}}(V_0)$ at Root Node

To evaluate $G_{\text{FOOTWEAR}}(V_0)$ at the root of a decision tree for the professor-student training set given in Fig. 3.1, in which the value of FOOTWEAR is missing for pattern T5, the following procedure is used. The root is named y_0 . The classes are C_1 (professor) and C_2 (student). Therefore, $m = 2$. Since FOOTWEAR has two possible values (clogs and sandals), root y_0 has two children y_1 and y_2 . Therefore, $n = 2$. To understand the following material easily, you may want to look at Fig. 3.2, since it examines FOOTWEAR at the root.

$V_0 = \{T1, T2, \dots, T8\}$ at node y_0

$Y(1, 0) = 3$ (no. of known FOOTWEAR-value patterns in V_0 of class C_1)

$Y(2, 0) = 4$ (no. of known FOOTWEAR-value patterns in V_0 of class C_2)

$Z(0) = 8$ (no. of patterns in V_0)

$Z'(0) = 7$ (no. of known FOOTWEAR-value patterns in V_0)

$V_1 = \{T1, T8\}$ at node y_1 , where FOOTWEAR = clogs

$Y(1, 1) = 0$ (no. of patterns in V_1 of class C_1)

$Y(2, 1) = 2$ (no. of patterns in V_1 of class C_2)

$Z(1) = 2$ (no. of patterns in V_1)

$V_2 = \{T2, T3, T4, T6, T7\}$ at node y_2 , where FOOTWEAR = sandals

$Y(1, 2) = 3$ (no. of patterns in V_2 of class C_1)

$Y(2, 2) = 2$ (no. of patterns in V_2 of class C_2)

$Z(2) = 5$ (no. of patterns in V_2)

$V_3 = \{T5\}$ (patterns with missing value of FOOTWEAR)

$Z(3) = 1$ (no. of patterns with missing value of FOOTWEAR)

$$I(V_0) = - \sum_{k=1}^m \frac{Y(k, 0)}{Z'(0)} \log \frac{Y(k, 0)}{Z'(0)} = 0.9852 \quad [\text{from Eqn (3.1)}]$$

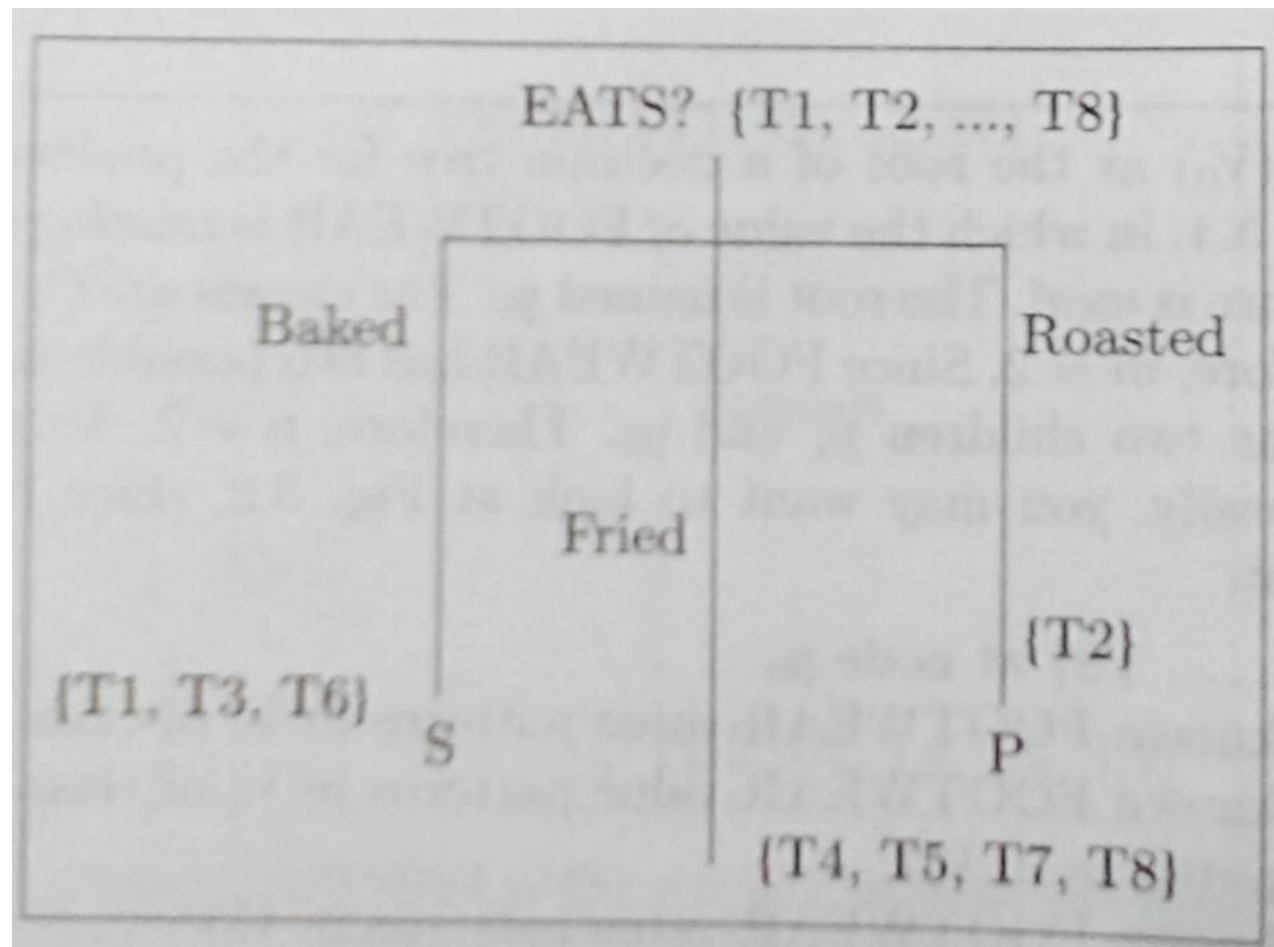
$$I_{\text{FOOTWEAR}}(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z'(0)} \sum_{k=1}^m \frac{Y(k, j)}{Z(j)} \log \frac{Y(k, j)}{Z(j)} = 0.6936 \quad [\text{from Eqn (3.2)}]$$

$$g_{\text{FOOTWEAR}}(V_0) = \frac{Z'(0)}{Z(0)} [I(V_0) - I_{\text{FOOTWEAR}}(V_0)] = 0.25515 \quad [\text{from Eqn (3.3)}]$$

$$S_{\text{FOOTWEAR}}(V_0) = - \sum_{j=1}^{n+1} \frac{Z(j)}{Z(0)} \log \frac{Z(j)}{Z(0)} = 1.3 \quad [\text{from Eqn (3.4)}]$$

$$G_{\text{FOOTWEAR}}(V_0) = \frac{g_{\text{FOOTWEAR}}(V_0)}{S_{\text{FOOTWEAR}}(V_0)} = 0.1963 \quad [\text{from Eqn (3.5)}]$$

Suppose Eats is at the Root Node and we calculate the Attribute for Next Node



Considering Footwear First

To evaluate $G_{\text{FOOTWEAR}}(V_0)$ at the middle child of the root of the decision tree given in Fig. 3.4, in which EATS has been examined at the root, the following procedure is used. The value of FOOTWEAR is missing for pattern T5. The middle child of the root is named y_0 . The classes are C_1 (professor) and C_2 (student). Therefore, $m = 2$. Since FOOTWEAR has two possible values (clogs and sandals), node y_0 has two children y_1 and y_2 . Therefore, $n = 2$.

$V_0 = \{T4, T5, T7, T8\}$ at node y_0

$Y(1, 0) = 2$ (no. of known FOOTWEAR-value patterns in V_0 of class C_1)

$Y(2, 0) = 1$ (no. of known FOOTWEAR-value patterns in V_0 of class C_2)

$Z(0) = 4$ (no. of patterns in V_0)

$Z'(0) = 3$ (no. of known FOOTWEAR-value patterns in V_0)

$V_1 = \{T8\}$ at node y_1 , where FOOTWEAR = clogs

$Y(1, 1) = 0$ (no. of patterns in V_1 of class C_1)

$Y(2, 1) = 1$ (no. of patterns in V_1 of class C_2)

$Z(1) = 1$ (no. of patterns in V_1)

$V_2 = \{T4, T7\}$ at node y_2 , where FOOTWEAR = sandals

$Y(1, 2) = 2$ (no. of patterns in V_2 of class C_1)

$Y(2, 2) = 0$ (no. of patterns in V_2 of class C_2)

$Z(2) = 2$ (no. of patterns in V_2)

$V_3 = \{T5\}$ (patterns with missing value of FOOTWEAR)

$Z(3) = 1$ (no. of patterns with missing value of FOOTWEAR)

$$I(V_0) = - \sum_{k=1}^m \frac{Y(k, 0)}{Z'(0)} \log \frac{Y(k, 0)}{Z'(0)} = 0.9183 \quad [\text{from Eqn (3.1)}]$$

$$I_{\text{FOOTWEAR}}(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z'(0)} \sum_{k=1}^m \frac{Y(k, j)}{Z(j)} \log \frac{Y(k, j)}{Z(j)} = 0 \quad [\text{from Eqn (3.2)}]$$

$$g_{\text{FOOTWEAR}}(V_0) = \frac{Z'(0)}{Z(0)} [I(V_0) - I_{\text{FOOTWEAR}}(V_0)] = 0.9183 \quad [\text{from Eqn (3.3)}]$$

$$S_{\text{FOOTWEAR}}(V_0) = - \sum_{j=1}^{n+1} \frac{Z(j)}{Z(0)} \log \frac{Z(j)}{Z(0)} = 1.4466 \quad [\text{from Eqn (3.4)}]$$

$$G_{\text{FOOTWEAR}}(V_0) = \frac{g_{\text{FOOTWEAR}}(V_0)}{S_{\text{FOOTWEAR}}(V_0)} = 0.6351 \quad [\text{from Eqn (3.5)}]$$

To evaluate $G_{\text{HABIT}}(V_0)$ at the middle child of the root of the decision tree given in Fig. 3.4, in which EATS has been examined at the root, the following procedure is used. The middle child of the root is named y_0 . The classes are C_1 (professor) and C_2 (student). Therefore, $m = 2$. Since HABIT has two possible values (gabby and quiet), node y_0 has two children y_1 and y_2 . Therefore, $n = 2$.

$V_0 = \{T4, T5, T7, T8\}$ at node y_0

$Y(1, 0) = 2$ (no. of patterns in V_0 of class C_1)

$Y(2, 0) = 2$ (no. of patterns in V_0 of class C_2)

$Z(0) = 4$ (no. of patterns in V_0)

$V_1 = \{T5, T7\}$ at node y_1 , where HABIT = gabby

$Y(1, 1) = 1$ (no. of patterns in V_1 of class C_1)

$Y(2, 1) = 1$ (no. of patterns in V_1 of class C_2)

$Z(1) = 2$ (no. of patterns in V_1)

$V_2 = \{T4, T8\}$ at node y_2 , where HABIT = quiet

$Y(1, 2) = 1$ (no. of patterns in V_2 of class C_1)

$Y(2, 2) = 1$ (no. of patterns in V_2 of class C_2)

$Z(2) = 2$ (number of patterns in V_2)

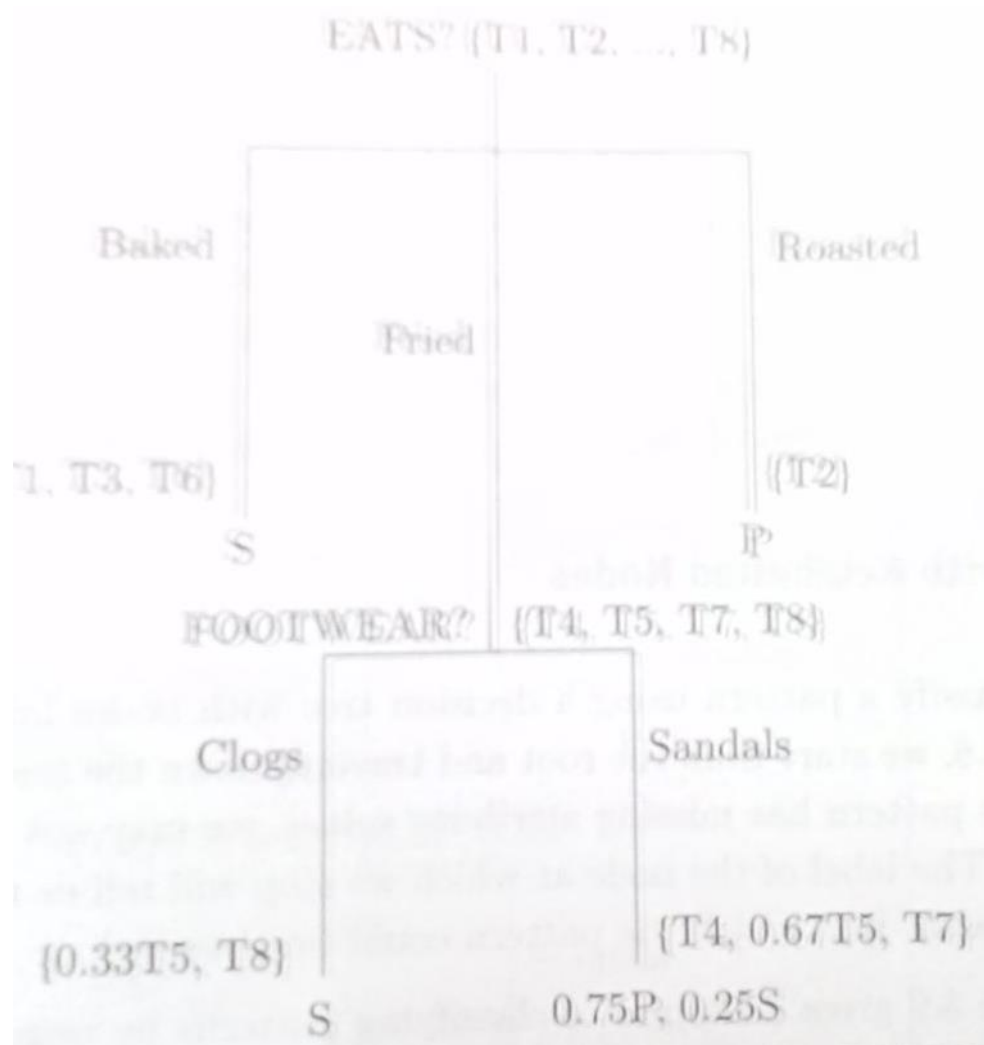
$$I(V_0) = - \sum_{k=1}^m \frac{Y(k, 0)}{Z(0)} \log \frac{Y(k, 0)}{Z(0)} = 1.0 \quad [\text{from Eqn (2.1)}]$$

$$I_{\text{HABIT}}(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z(0)} \sum_{k=1}^m \frac{Y(k, j)}{Z(j)} \log \frac{Y(k, j)}{Z(j)} = 1.0 \quad [\text{from Eqn (2.2)}]$$

$$S_{\text{HABIT}}(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z(0)} \log \frac{Z(j)}{Z(0)} = 1.0 \quad [\text{from Eqn (2.4)}]$$

$$G_{\text{HABIT}}(V_0) = \frac{I(V_0) - I_{\text{HABIT}}(V_0)}{S_{\text{HABIT}}(V_0)} = 0 \quad [\text{from Eqn (2.5)}]$$

Decision Tree After Considering Attributes



Weights to Leaves

2. The sum of the weights is calculated as

$$\text{sum_of_weights} = \sum_{i=1}^n w_i$$

3. For every training pattern T in V_0 whose value of attribute A is missing, do steps 3.1 and 3.2.

3.1. Let f_0 be the fraction of pattern T present in V_0 .

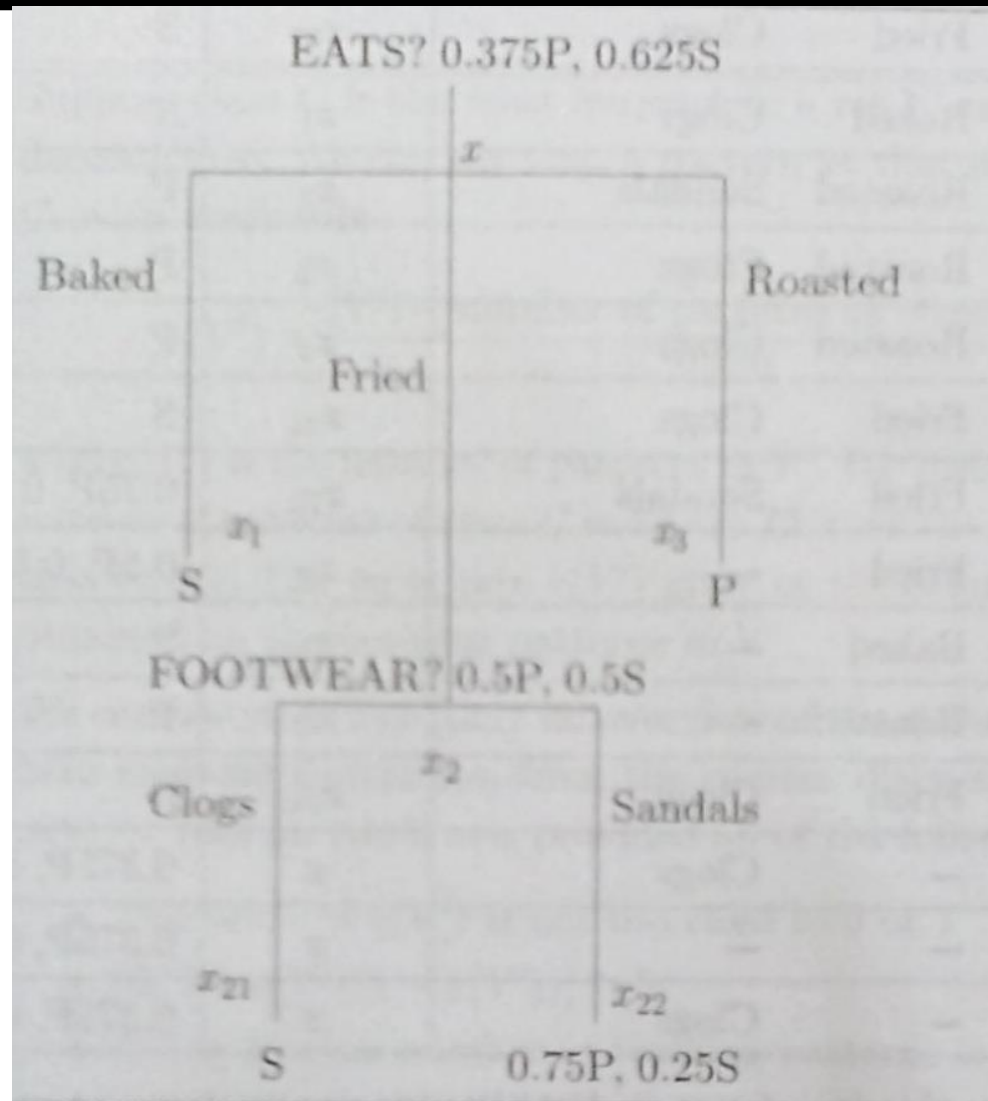
3.2. For $j = 1, 2, \dots, n$ do step 3.2.1.

3.2.1. Add to set V_j the f_j th fraction of pattern T , where

$$f_j = \frac{w_j}{\text{sum_of_weights}} f_0$$

4. Return from the procedure.

Final Decision Tree



Classification of Patterns

T1	Gabby	Baked	Clogs	x_1	S
T2	Gabby	Roasted	Sandals	x_3	P
T3	Gabby	Baked	Sandals	x_1	S
T4*	Quiet	Fried	Sandals	x_{22}	0.75P, 0.25S
T5*	Gabby	Fried	—	x_2	0.5P, 0.5S
T6	Quiet	Baked	Sandals	x_1	S
T7*	Gabby	Fried	Sandals	x_{22}	0.75P, 0.25S
T8	Quiet	Fried	Clogs	x_{21}	S
R1	Quiet	Baked	Clogs	x_1	S
R2	Quiet	Roasted	Sandals	x_3	P
R3	Gabby	Roasted	Clogs	x_3	P
R4	Quiet	Roasted	Clogs	x_3	P
R5	Gabby	Fried	Clogs	x_{21}	S
R6	—	Fried	Sandals	x_{22}	0.75P, 0.25S
R7	—	Fried	—	x_2	0.5P, 0.5S
R8	—	Baked	—	x_1	S
R9	—	Roasted	—	x_3	P
R10	—	Fried	Clogs	x_{21}	S
R11	—	—	Clogs	x	0.375P, 0.625S
R12	Gabby	—	—	x	0.375P, 0.625S
R13	Gabby	—	Clogs	x	0.375P, 0.625S

FIGURE 3.9 Classifications of some professor-student patterns using the decision tree given in Fig. 3.1. A missing value of an attribute is shown by a dash. Patterns T1 to T8 are the training patterns taken from Fig. 3.1. An asterisk besides the name of a training pattern indicates that its classification here is not identical to that in its training set given in Fig. 3.1. Patterns R1 to R13 are recall patterns, of which R1 to R4 are the same as those in Fig. 3.1. The classifications for patterns R11 to R13 have been obtained from the label at the root of the decision tree. For such patterns, an alternative method of classification is discussed in Section 3.2.

Activities

- Read Chapters 2 and 3 from Shinghal, (2011)
- Do Assignment 2 – coming very soon!



Questions or Comments?